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- This talk is about plasma turbulence and why it is the dominant limitation to magnetic confinement fusion.
- The focus is not on turbulence modeling itself, but on how we interrogate experimental fluctuation data.
- In particular, I will discuss nonlinear, directional correlation methods developed to go beyond standard diagnostics.

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- Fusion research originates from early nuclear physics, beginning with Rutherford's first fusion reaction in 1919.
- By the 1920s, mass defect and nuclear binding energy explained how fusion could release enormous energy.
- Eddington proposed that stellar energy originates from hydrogen fusion.
- In 1938, Bethe and Weizscker established the proton-proton chain and CNO cycle.
- In the 1950s, fusion research accelerated through weapons programs and the invention of toroidal confinement concepts.
- The 1968 Soviet T-3 tokamak demonstrated dramatically improved confinement, launching modern tokamak research.
- Since the 1990s, the primary challenge has not been ignition physics, but turbulence and confinement degradation.
- Modern devices like ITER and EAST are fundamentally limited by turbulence-driven transport.

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- Plasma is the fourth state of matter, consisting of charged particles interacting through Coulomb forces.
- Unlike neutral fluids, plasma dynamics are strongly constrained by magnetic fields.
- A tokamak confines plasma magnetically in a toroidal geometry to avoid material walls.
- Charged particles spiral along magnetic field lines while slowly drifting across them.

- Toroidal and poloidal magnetic fields combine to form closed helical particle orbits.
- Ideal confinement would restrict transport across magnetic flux surfaces.

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- Experimental transport levels exceed classical and neoclassical predictions by one to two orders of magnitude.
- This discrepancy is known as anomalous transport.
- Turbulence arises from drift-type instabilities driven by pressure gradients and electric fields.
- Plasma turbulence is anisotropic, intermittent, and non-Gaussian.
- Zonal flows represent a form of self-organization analogous to atmospheric bands on Jupiter.
- Modern transport analysis reveals both slow diffusive transport and fast nonlocal transport events.
- These fast events resemble Levy flights rather than standard random walks.

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- Experimental turbulence measurements are typically obtained using Langmuir probes.
- Floating potential fluctuations provide access to local electrostatic turbulence.
- Multiple spatially separated probes allow the study of radial and poloidal correlations.
- The experiments referenced here follow setups used by Bsharat et al. and Alipour et al.
- These measurements provide time series suitable for nonlinear statistical analysis.

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- Conventional analysis tools include linear cross-correlation, FFTs, and coherence.
- These methods assume Gaussian statistics and waveform coherence.

- Plasma turbulence violates these assumptions due to shear, intermittency, and deformation.
- Linear correlation is symmetric and cannot identify causal or directional relationships.
- This motivates the use of nonlinear, phase-space-based techniques.

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- Ding et al. introduced a nonlinear cross-correlation method based on conditional dispersion.
- The method reconstructs local dynamics using delay-coordinate embedding.
- An asymmetry coefficient S_{xy} quantifies directional coupling between signals.
- If $S_{xy} > 0$, energy or information flows from x to y .
- van Milligen et al. use Transfer Entropy, an information-theoretic measure of causality.
- Transfer Entropy quantifies how much the past of one signal predicts the future of another.
- Both methods aim to extract directional information flow in nonlinear systems.

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- Time series are embedded into phase space using delay vectors.
- The embedding dimension M and delay m are chosen using autocorrelation times.
- Conditional dispersion measures how spread out one attractor is when the other is locally constrained.
- A neighborhood size ϵ is swept to construct dispersion curves.
- The nonlinear correlation strength g_{xy} is extracted from ϵ_{90} .
- Directionality is determined using the asymmetry coefficient S_{xy} .

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- The algorithm is first validated using coupled Van der Pol oscillators.
- The direction of coupling is known a priori in the synthetic system.
- This provides a controlled benchmark for the nonlinear method.
- The method is then applied to experimental HT-7 tokamak data.
- Results are compared directly against Dings original MATLAB implementation.

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- Dispersion curves for the Van der Pol system recover the correct coupling direction.
- HT-7 dispersion curves match previously published results.
- Nonlinear correlation lengths are significantly larger than linear correlation lengths.
- Direction reversals are observed across confinement transitions.
- These results support the physical interpretation of inward and outward turbulence propagation.

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- Fully reproduce Dings algorithm in a modern, open-source language.
- Validate results against synthetic and experimental data sets.
- Establish mathematical links between NLCC, Transfer Entropy, and CTRW models.
- Interpret experimental results within a stochastic transport framework.
- Quantify the relative contribution of fast and slow transport channels.
- Ultimately connect directional turbulence propagation to confinement degradation.